Conveyor Control Simulation using Siemens S7-1200
TIA Portal V20 | S7-PLCSIM V20 | Ladder Logic

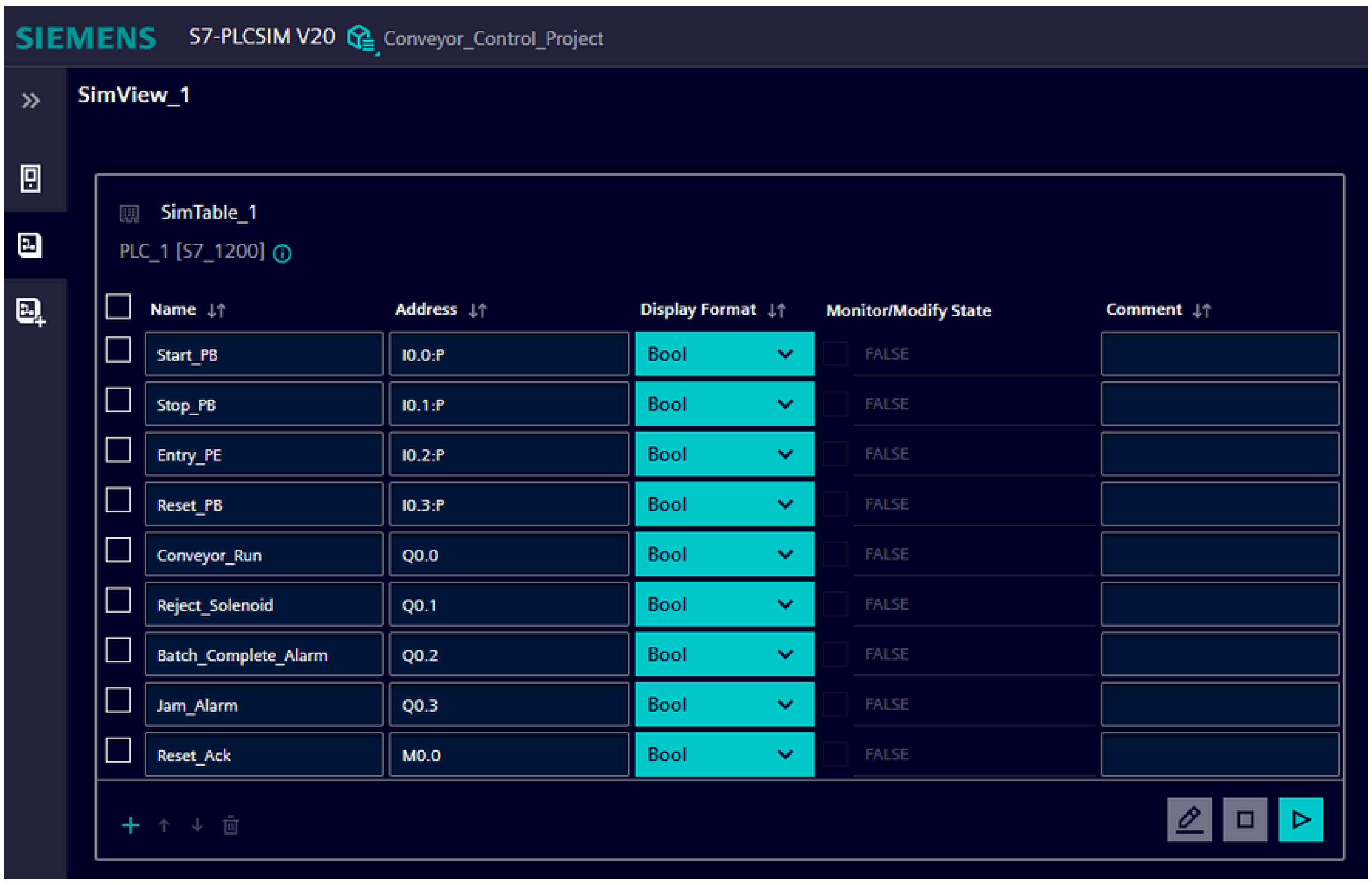
Functions implemented:
- Motor start/stop latch
- Product counting using CTU counter
- Conveyor stop at batch count = 5
- Reject output activation
- Batch complete alarm
- Timer-based jam alarm
- Reset acknowledgment

Work Flow:

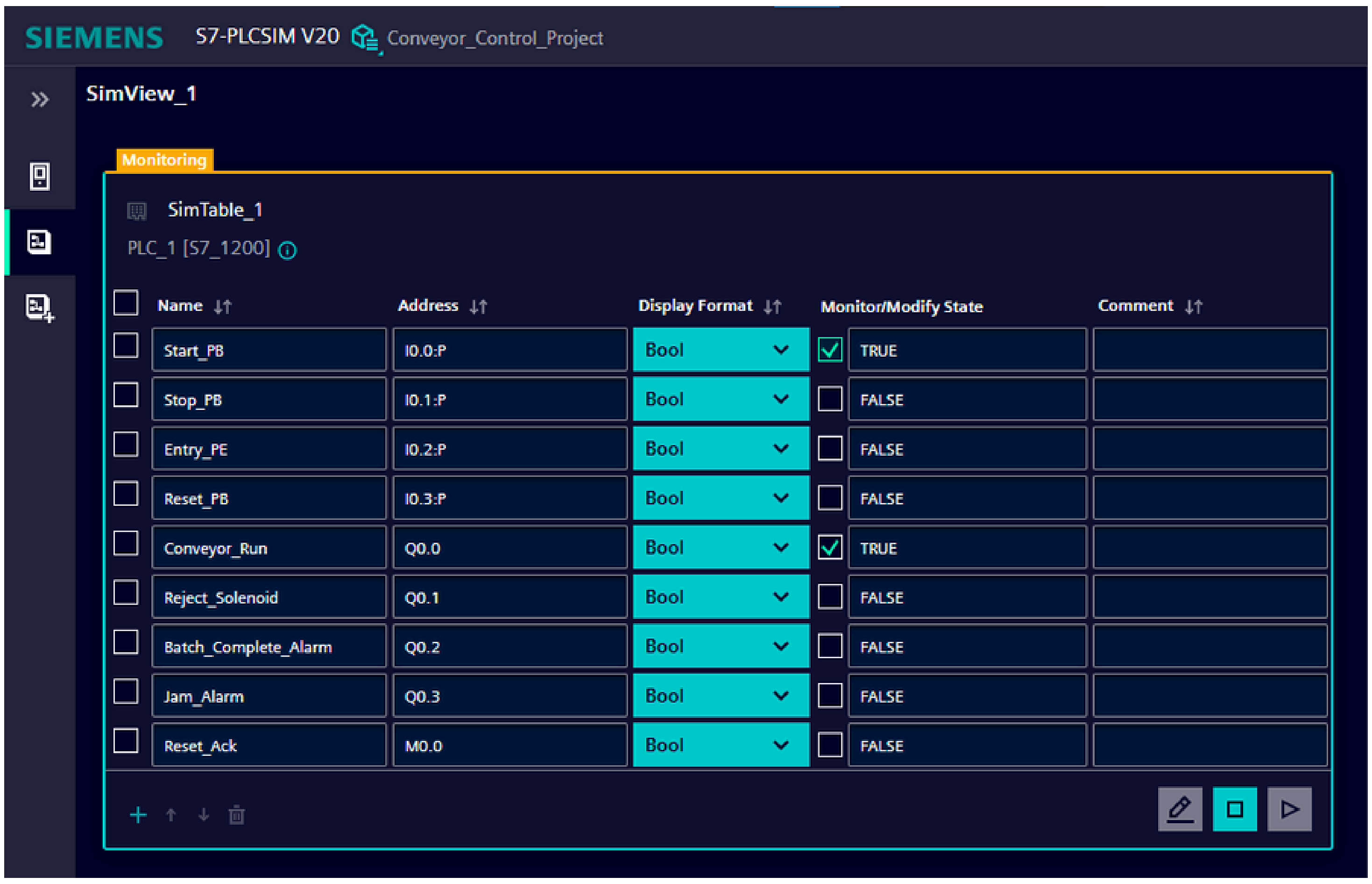
Start → Conveyor Runs → Product Count → Count = 5 → Conveyor Stops → Alarm ON → Reset

Step 1 - Idle State and Start Verification

Initial idle condition verified before cycle execution.

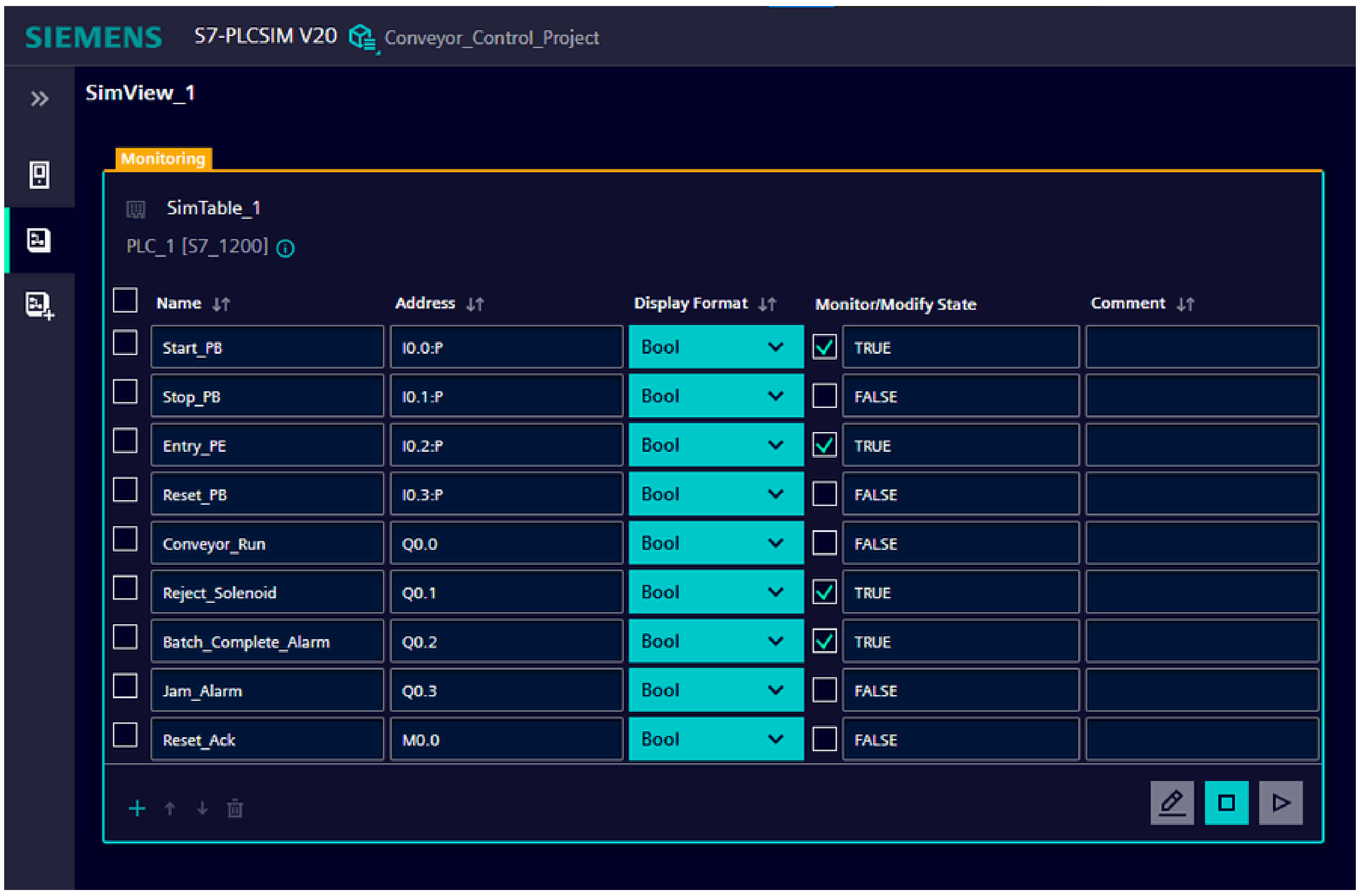


Motor latch activated and conveyor running.

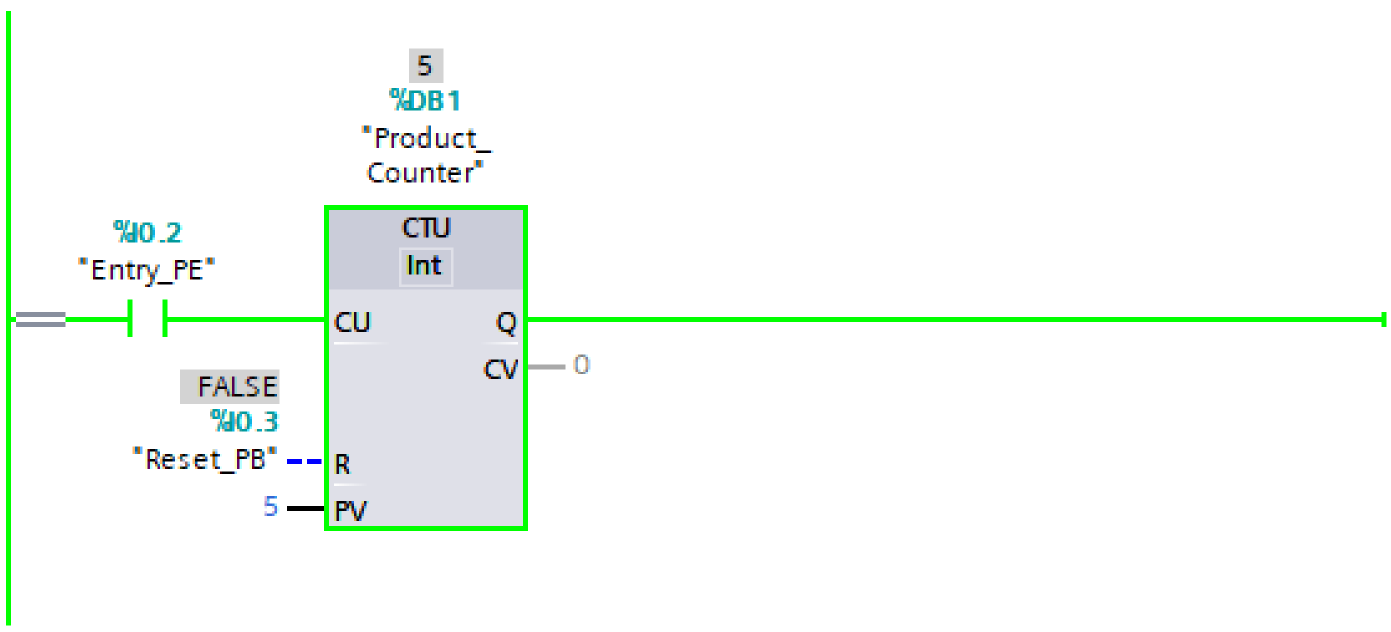


Step 2 - Product Counting and Batch Completion

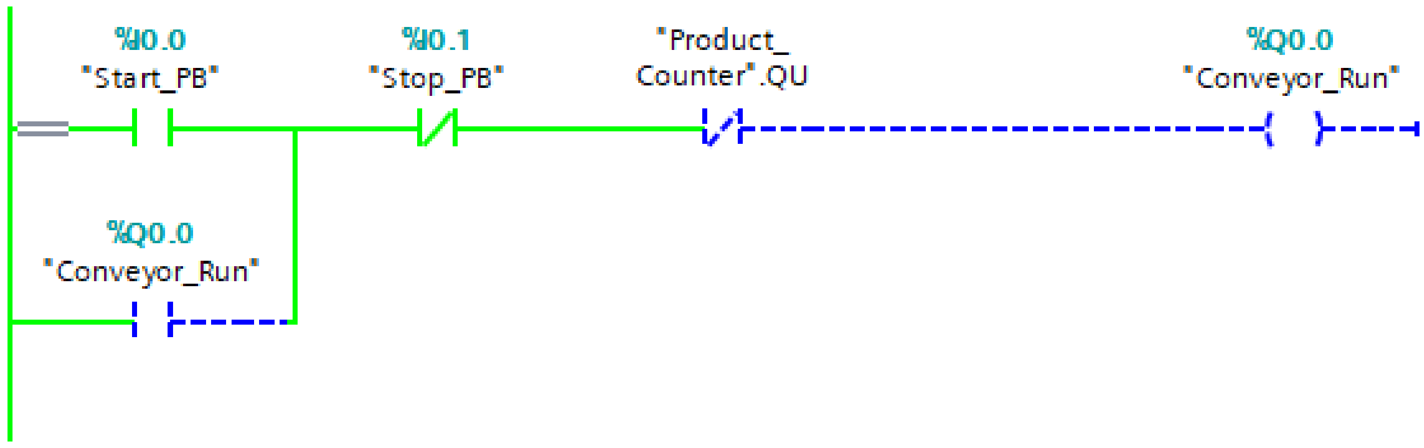
Counter reached preset value after entry pulses (CV = 5)



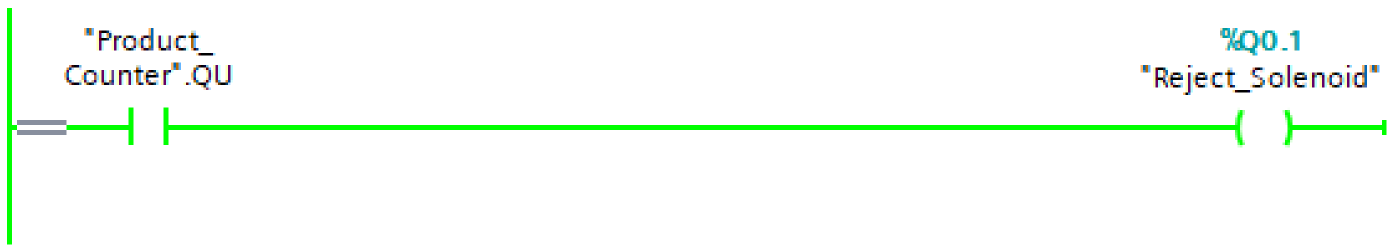
Name	Address	Display format	Monitor value
"Start_PB"	%I0.0	Bool	TRUE
"Stop_PB"	%I0.1	Bool	FALSE
"Conveyor_Run"	%Q0.0	Bool	FALSE
"Entry_PE"	%I0.2	Bool	TRUE
"Reset_PB"	%I0.3	Bool	FALSE
"Product_Counter".QU		Bool	TRUE
"Product_Counter".CV		DEC+/-	5
"Reject_Solenoid"	%Q0.1	Bool	TRUE
"Batch_Complete_...	%Q0.2	Bool	TRUE
"Jam_Alarm"	%Q0.3	Bool	FALSE
"Reset_Ack"	%M0.0	Bool	FALSE



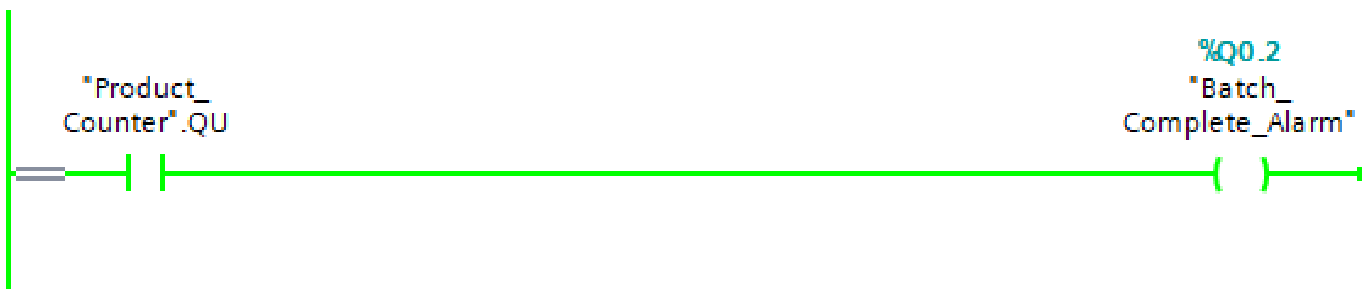
Conveyor OFF



Reject output ON

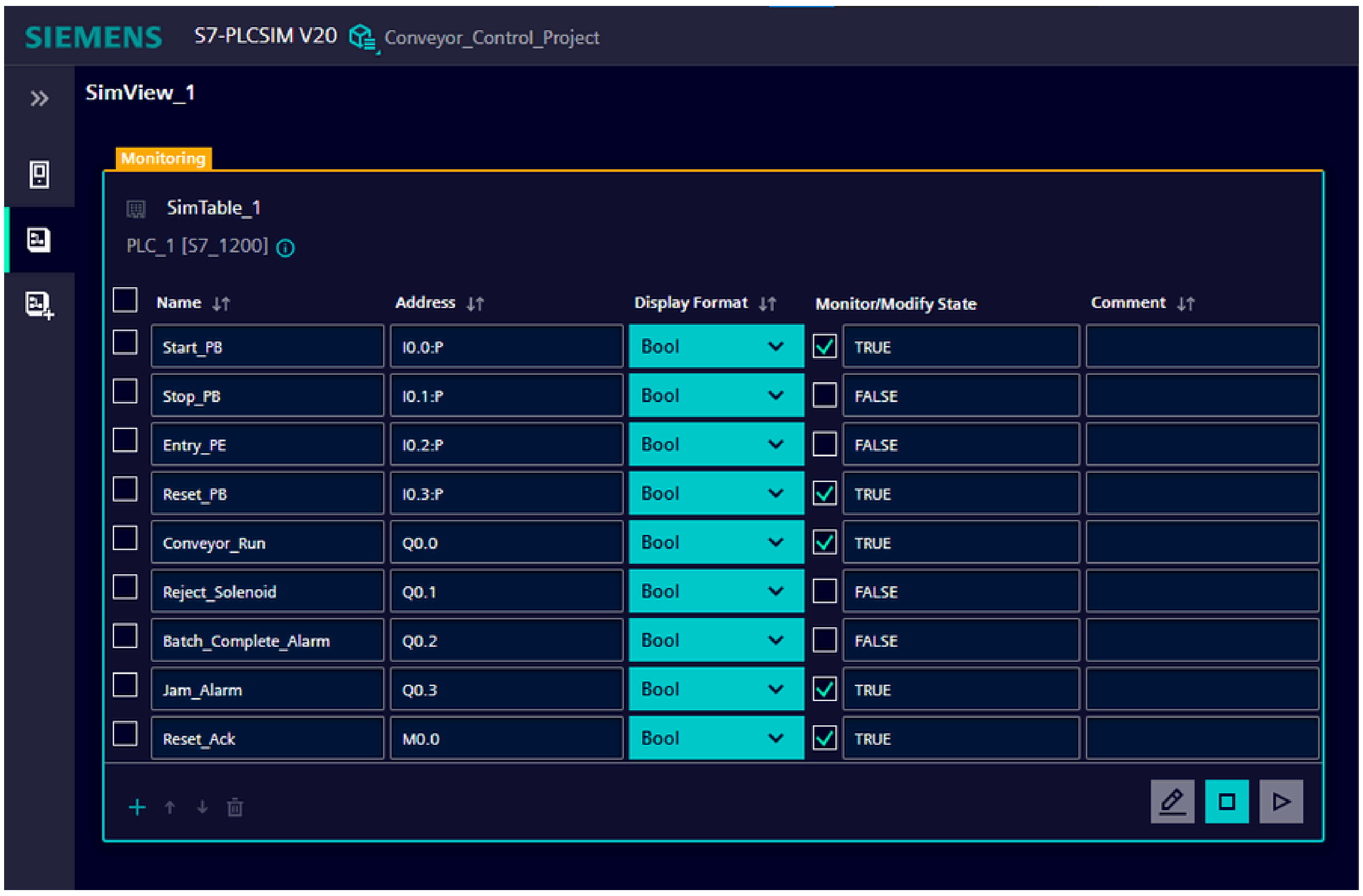


Batch alarm ON



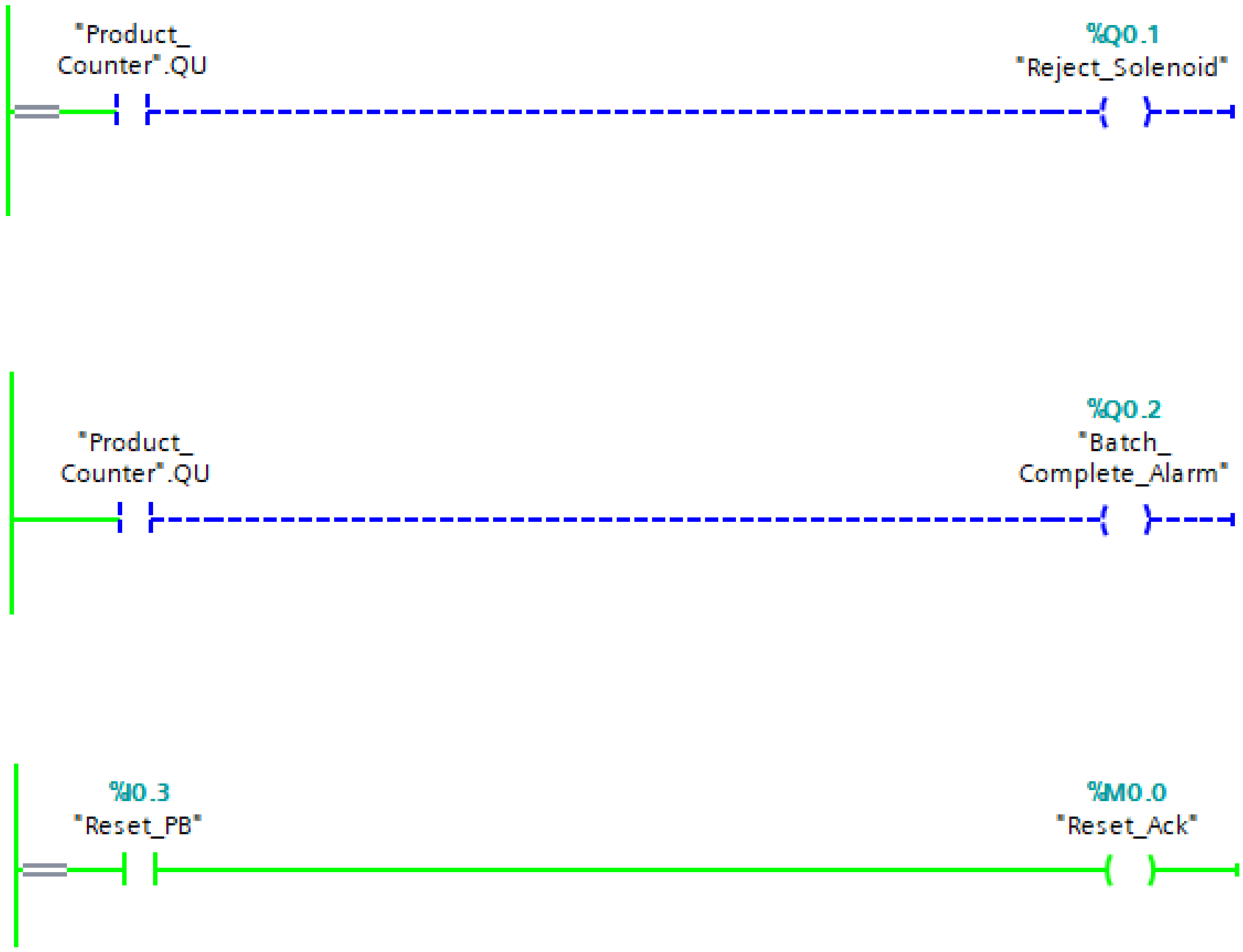
Step 3 - Reset and System Recovery

Reset command applied before output recovery



Name	Address	Display format	Monitor value
"Start_PB"	%I0.0	Bool	TRUE
"Stop_PB"	%I0.1	Bool	FALSE
"Conveyor_Run"	%Q0.0	Bool	TRUE
"Entry_PE"	%I0.2	Bool	TRUE
"Reset_PB"	%I0.3	Bool	FALSE
"Product_Counter".QU		Bool	FALSE
"Product_Counter".CV		DEC+/-	0
"Reject_Solenoid"	%Q0.1	Bool	FALSE
"Batch_Complete_...	%Q0.2	Bool	FALSE
"Jam_Alarm"	%Q0.3	Bool	TRUE
"Reset_Ack"	%M0.0	Bool	FALSE

Reject and batch outputs OFF
Reset acknowledgment active during reset command



Final ladder state after sequence verification.

Project Validation Summary
✓ **Motor latch verified**
✓ **Counter reached preset value**
✓ **Conveyor stopped automatically**
✓ **Reject and batch alarm triggered**
✓ **Timer alarm verified**
✓ **Reset restored system state**
Simulation Environment:
TIA Portal V20 + S7-PLCSIM V20

Conveyor_Control_Project / PLC_1 [CPU 1214C DC/DC/DC] / Program blocks

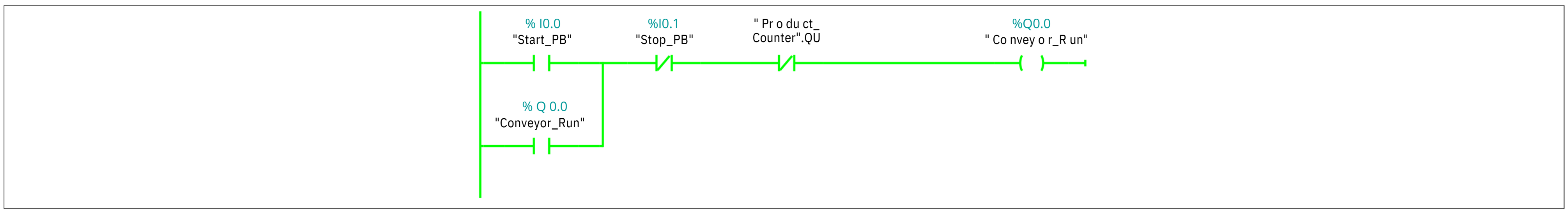
Main [OB1]

Main Properties

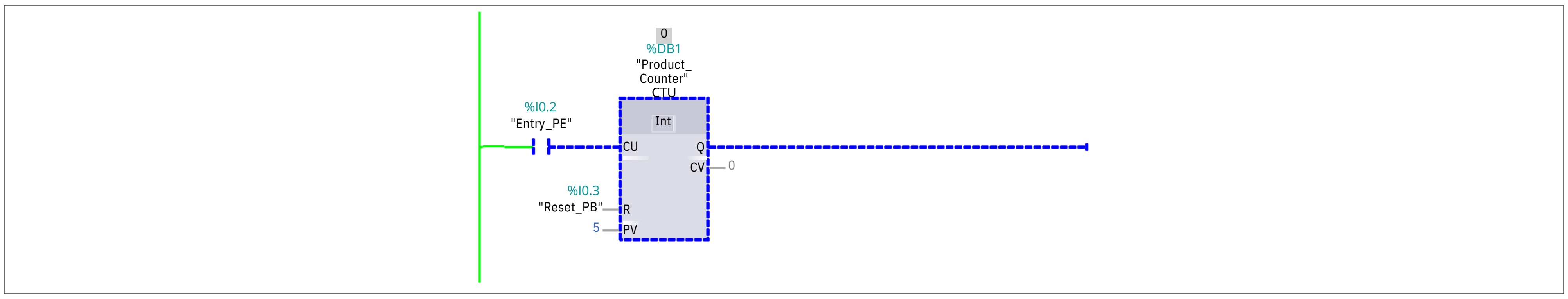
General							
Name							
Numbering	Main	Number	1	Type	OB	Language	LAD
Information	A utomatic						
Title							
	"Main Program Sweep (Cy-cle)"	A utho r		Co mment		Family	
Version	0.1	User-defined ID					

Main			
Name	Data type	Default value	Comment
Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

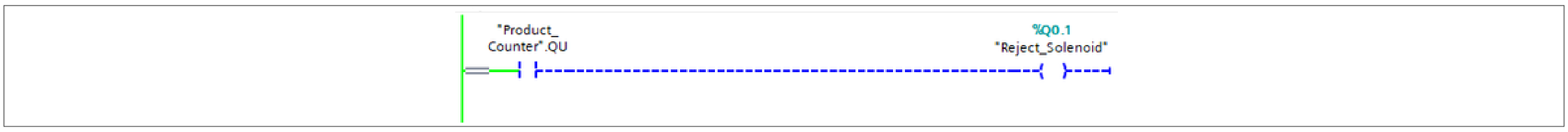
Network 1: Motor Latch



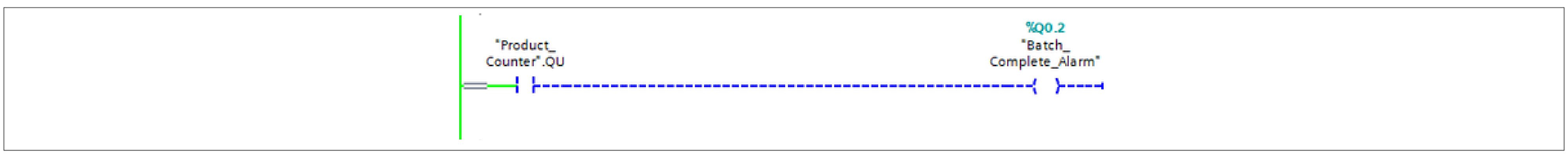
Network 2: Counter



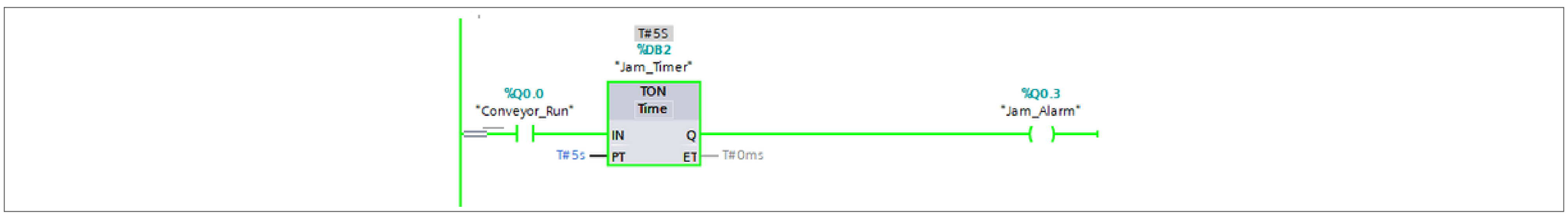
Network 3: Reject



Network 4: Batch alarm



Network 5: Timer jam logic



Network 6: Reset acknowledgment